Controllable Chord Progression Reharmonization via RVAE-LSTM using Perceptually-Grounded Complexity Score

A model that changes a chord progression to a chosen level of complexity.

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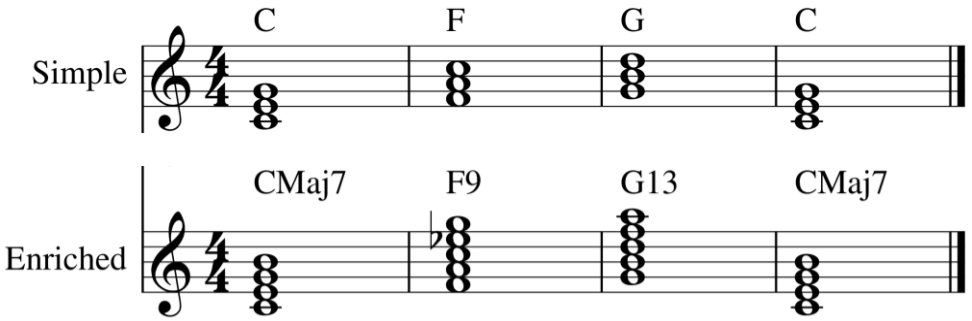
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Motivation

Richer chords simply have more notes: compare these two versions

- **Chord:** 3+ notes played together (C–E–G).
- **Progression:** a series of chords in order.
- **Richer chord:** more notes (C–E–G–B).
- **Reharmonization:** changing the chords.
- **The gap:** no tool lets you choose how much.



Hand-written example. Simple: 3 notes per chord. Enriched: 4–6 notes per chord.

What Others Did and What Is Missing

Can a model target what listeners perceive, one property at a time?

Earlier systems

- **Comanducci et al. [2]:** one complexity number from a statistical model. 4 chord types; synthetic 5-chord loops in C major/minor.
- **Uemura et al. [6]:** moves chords inside a trained model; no target level.
- **MusicVAE [5]:** generates music; not for controlled enrichment.

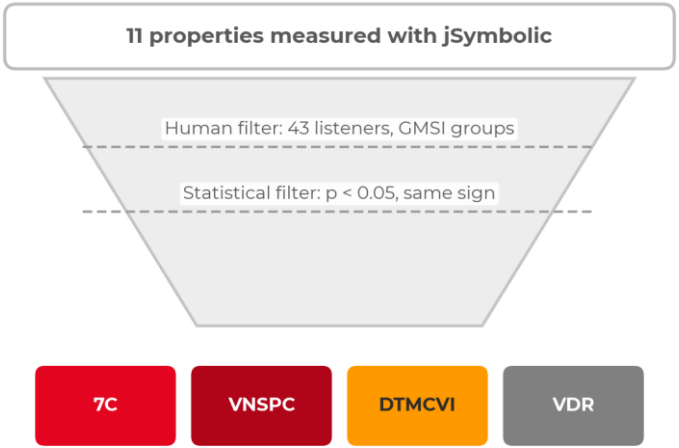
The gap

	Target	Property	Listener score
Comanducci [2]	5 levels	No	No
Uemura [6]	No	No	No
MusicVAE [5]	No	No	No
PEARL	Any	**Yes**	**Yes**

No earlier system offers all three.

A Complexity Score from Listeners (PCS)

A human filter and a statistical filter select four properties



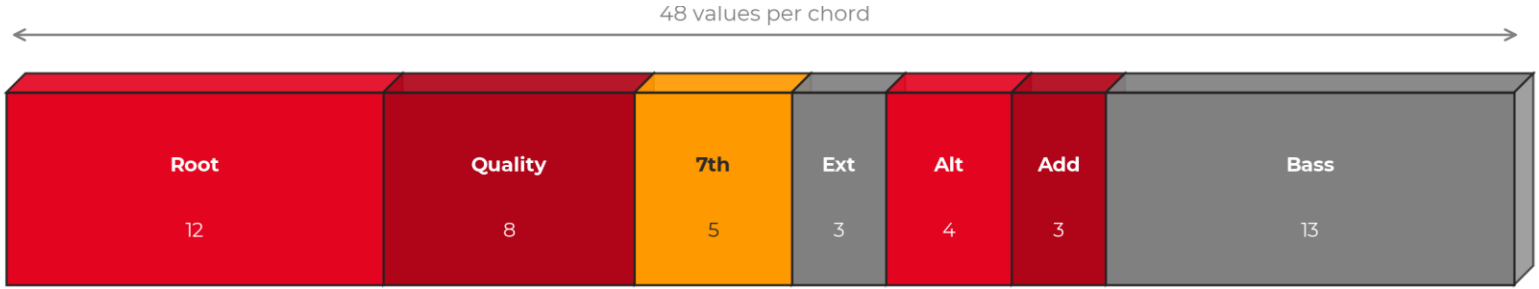
Property	Meaning	Weight
Seventh chords	Share of chords with a seventh	0.443
Notes per chord	Variation in simultaneous note count	0.378
Interval spacing	Distance between the two common intervals	0.338
Dissonance	Share of dissonant vertical intervals	0.201

v1 uses these weights; v2 weights all four equally. Seventh chords weigh most.

53 joined; 43 valid; 1,032 pairwise comparisons.

How a Chord Becomes 48 Numbers

Seven slots, each with a musical meaning



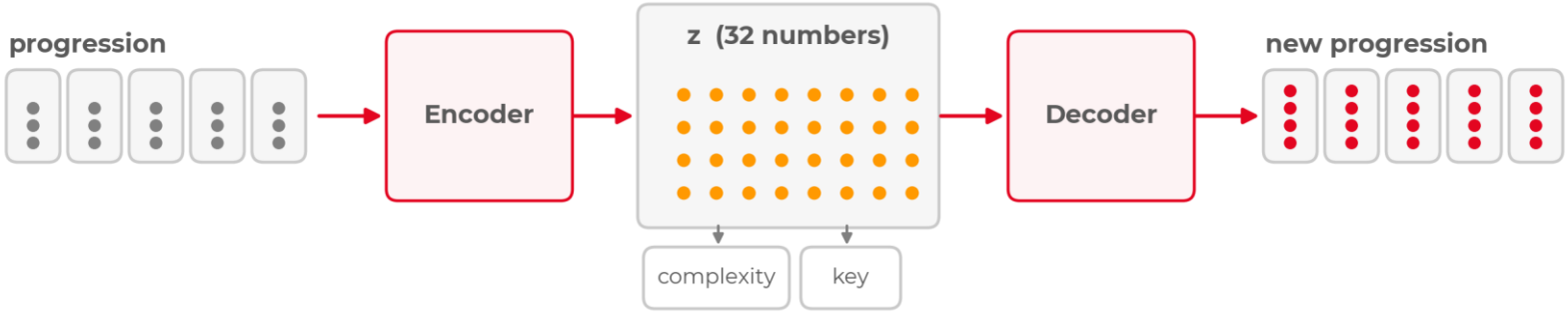
Example: Cmaj7



4 active values: root C · quality major · seventh maj7 · bass root

The PEARL Model

A network that turns a progression into 32 numbers, then rebuilds it [2, 5]



- **Encoder:** a progression → 32 numbers (z).
- **Decoder:** 32 numbers → a progression.
- **Two predictors:** one reads complexity, one reads key.
- **We move z to change complexity;** the key stays readable.

Reaching a Target Complexity

The space is ordered by complexity; we move z toward the target

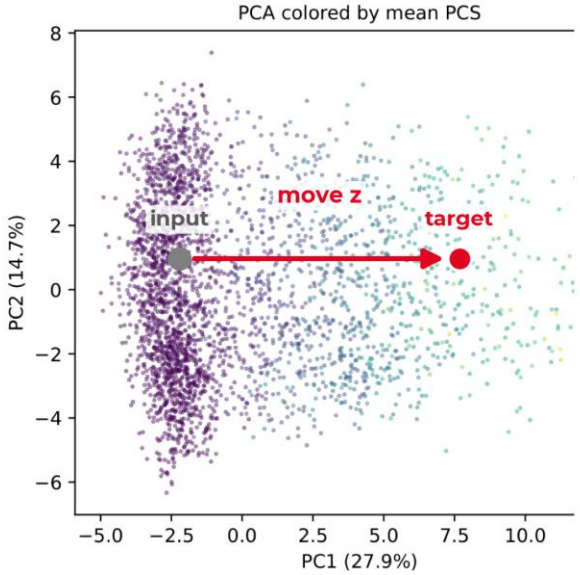
- **Goal:** move z until complexity = target.
- About 20–30 small steps.
- A penalty keeps z close to the original.
- **One property at a time.**

Example: seventh chords, target 1.0

Input
Cmin – Gmaj – Cmin – Fmin – Cmin

Output
Cmaj7 – Dmin7 – Cmin7 – Gmin7 – Cmin7

Any target in [0, 1]; no retraining.



Experimental Setup

Data, training and baselines

DATA

877k

progressions

real songs [1]

SPLIT

80/10/10

by song

results on unseen songs

TASK

5

targets

from 0.00 to 1.00

COMPARE

2

baselines

CVAE · RVAE [2]

Every model gets the same input loop and the same five targets.

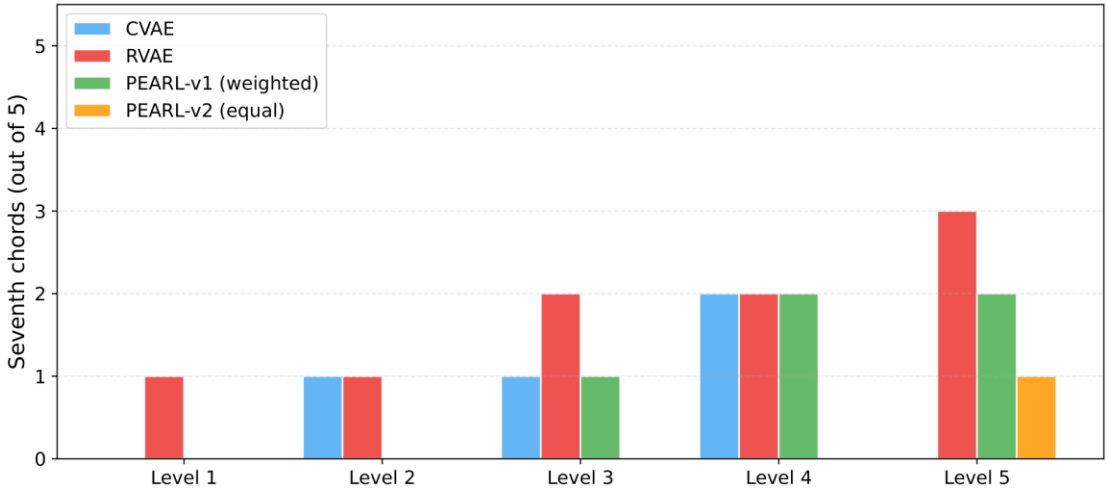
Evaluation Metrics

The score comes from listeners, not statistics. These metrics check the target and the key

Metric	What it measures
Target reached	Complexity of the output vs the target.
Property changed	Change in each of the four properties.
Stays in key	Share of roots in the original key [7].
Keeps roots	Share of positions with the same root.
Keeps quality	Share of chords with the same major/minor quality.
Distance moved	How far z moved from the original.

Results: Five Complexity Levels

How many seventh chords each model uses at each level



- **Both versions** reach the target in 20–30 steps.
- **v1:** about 1/5 of the change is seventh chords; v2: 1/10.
- **v2** spreads the change over the four properties.

Baselines use a different scale; seventh-chord counts still show how much each model enriches.

Result: Choose What Changes

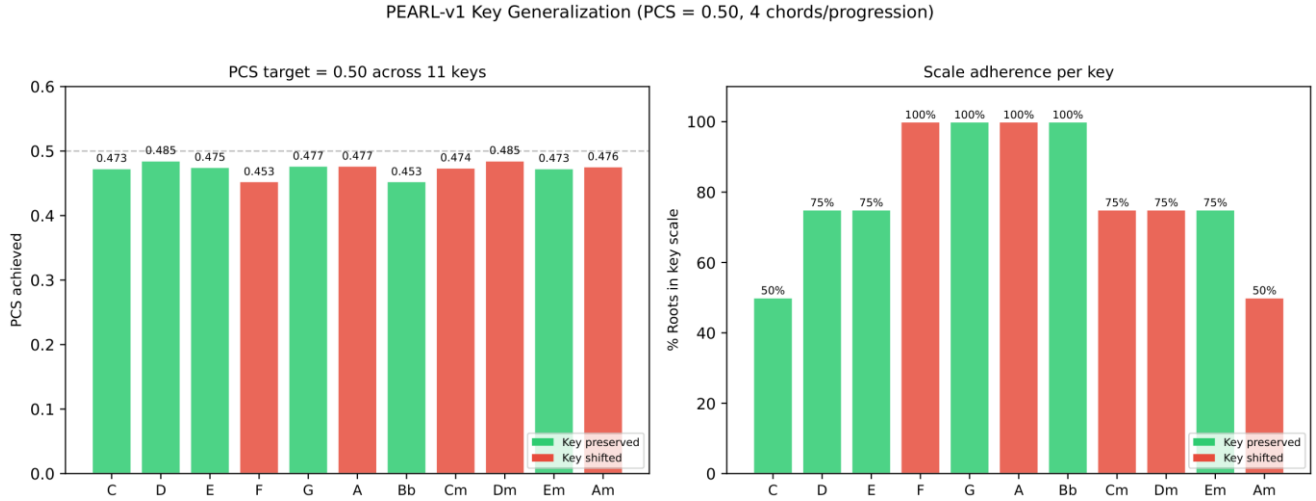
Earlier systems return one number; PEARL can target one property

- **Seventh chords only:** score 0.002 → 0.982; all five chords get a seventh.
- **Dissonance only:** 0.005 → 0.987; no seventh chords added.
- **Note variety only:** −0.002 → 0.983; other properties unchanged.

Seventh-chord target	**Cmaj7** 4 notes	**Dmin7** 4 notes	**Cmin7** 4 notes	**Gmin7** 4 notes	**Cmin7** 4 notes
Dissonance target	**Dmin** 3 notes	**Amaj** 3 notes	**Csus4add9** 4 notes	**Dsus4** 3 notes	**Amin** 3 notes

Results: Other Keys and Lengths

11 keys (7 major, 4 minor) and 4–12 chords, target PCS = 0.50



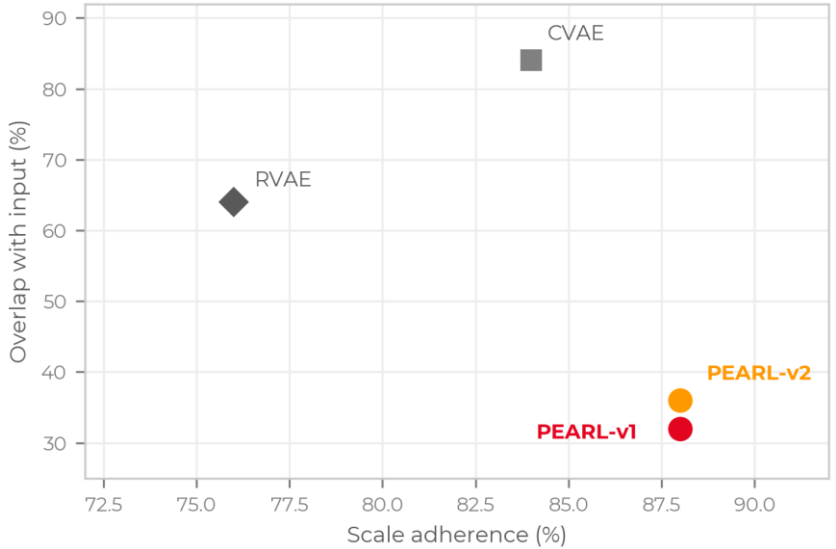
- **All 11 keys** reach the target (0.45–0.49).
- **80% of roots** stay in key; 100% at every length.
- **Key kept in 6 of 11;** the others move to a related key.

Results: Staying in Key

Five levels, input in C minor

- **88% of roots stay in key** vs CVAE 84% and RVAE 76% [2].
- **Overlap 32–36%** vs 84% for CVAE: PEARL changes more chords.
- **Low overlap is expected:** the anchor favours the target.

Mean of five levels, input in C minor. Overlap = same root at the same position.



Conclusions

Summary, limitations and future work

- **Score:** from listeners, not statistics.
- **Control:** any target in under 30 steps, globally or per property.
- **Key:** 88% of roots stay in key, across 11 keys and 4–12 chords.
- **Limits:** harmony only; 43 listeners; key kept in 6 of 11.
- **Next:** user-controlled penalty, rhythm, larger studies.

Code, checkpoint, audio/MIDI and survey data: github.com/Pepebeats-flp/pearl-sc

Thank you — Questions?

PEARL — Controllable Chord Progression Reharmonization

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github.com/Pepebeats-flp/pearl-sc

References

Works cited in this talk — full details in the paper

[1] S. Kantarelis et al., “Chordonomicon: A Dataset of 666,000 Songs and Their Chord Progressions,” arXiv:2410.22046, 2024.

[2] L. Comanducci, D. Gioiosa, M. Zanoni, F. Antonacci and A. Sarti, “Variational Autoencoders for Chord Sequence Generation Conditioned on Western Harmonic Music Complexity,” EURASIP J. Audio Speech Music Process., 2023.

[3] C. McKay, J. Cumming and I. Fujinaga, “jSymbolic 2.2: Extracting Features from Symbolic Music for Use in Musicological and MIR Research,” in Proc. ISMIR, 2018.

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[6] A. Uemura and T. Kitahara, “Morphing-Based Reharmonization with VAE: Reducing Dissonance with Consonance-Based Loss Function,” in Proc. AIMC, 2022.

[7] D. Temperley, The Cognition of Basic Musical Structures. MIT Press, 2004.